

MATH 180 Graph Theory — Discussion 10

Indicator variables and linearity of expectation (Ch. 10.3)

Quick Recall (Ch. 10.3)

Random variable & expectation. Let (Ω, P) be a finite probability space. A *random variable* is a function $f : \Omega \rightarrow \mathbb{R}$. Its *expectation* is

$$\mathbb{E}[f] = \sum_{\omega \in \Omega} P(\{\omega\}) \cdot f(\omega).$$

When P is uniform on Ω , this is just the average value $\mathbb{E}[f] = \frac{1}{|\Omega|} \sum_{\omega} f(\omega)$.

Indicator of an event. For an event $A \subseteq \Omega$, the *indicator* I_A is the random variable

$$I_A(\omega) = \begin{cases} 1 & \omega \in A, \\ 0 & \omega \notin A. \end{cases}$$

Fact. $\mathbb{E}[I_A] = P(A)$.

Linearity of expectation. For random variables f, g on the same (Ω, P) and any $\alpha \in \mathbb{R}$,

$$\mathbb{E}[\alpha f + g] = \alpha \mathbb{E}[f] + \mathbb{E}[g].$$

Independence is not required.

Random coin-toss space C_n . $\Omega = \{0, 1\}^n$ with the uniform distribution, modeling n independent fair coin tosses (write 0 = T, 1 = H).

Mini-example (lecture). Let $f_1 : C_n \rightarrow \mathbb{R}$ count the number of 1's. Setting $A_i = \{s \in C_n : s_i = 1\}$, we have $f_1 = \sum_{i=1}^n I_{A_i}$ and $P(A_i) = 1/2$, so by linearity

$$\mathbb{E}[f_1] = \sum_{i=1}^n P(A_i) = \frac{n}{2}.$$

Practice: Textbook Exercise 10.3.6*

Exercise 10.3.6*. We toss a fair coin n times. What is the expected number of “runs”? Runs are consecutive tosses with the same result. For instance, the toss sequence HHHTTHTH has 5 runs.